A. Code and Data Files

This appendix summarizes the MATLAB and Python scripts used for signal generation, model training, validation, and testing, along with their corresponding input and output files. Due to submission size limitations, some files were excluded from the submission.

A. Signal Generation

1. Constant-Frequency Signals
- **Script:** `signals_generator_const.m`
- **Output files:**
  - `train_signals_const.csv`
  - `val_signals_const.csv`
  - `test_signals_const.csv`

2. Time-Varying Frequency Signals
- **Script:** `signals_generator_varying.m`
- **Output files:**
  - `train_signals_varying.csv`
  - `val_signals_varying.csv`
  - `test_signals_varying.csv`
  - `train_metadata.csv`
  - `val_metadata.csv`
  - `test_metadata.csv`

B. Real Experimental Data Preprocessing

The data were obtained from the publicly available repository:

https://data.mendeley.com/datasets/v43hmbwxpm/2
- **Script:** `real_signals_preprocessing.m`
- **Input files:**
  - `H-A-1.mat`, `H-A-2.mat`, `H-A-3.mat`
  - `H-B-1.mat`, `H-B-2.mat`, `H-B-3.mat`
- **Output file:** `test_real_dataset.csv`

C. Model Training and Validation

1. Original MLP: Constant-Frequency Signals
- **Script:** `train_validate_const_mlp.py`
- **Input data:**
  - `train_signals_const.csv`
  - `val_signals_const.csv`
- **Output directory:** `results_const_mlp/`
- **Output files:**
  - `best_frequency_const_mlp.pth`
  - `training_history_const_mlp.csv`
  - `validation_results_const_mlp.csv`
  - `training_validation_loss_const_mlp.png`

2. Modified MLP: Constant-Frequency Signals
- **Script:** `train_validate_const_modified_mlp.py`
- **Input data:**
  - `train_signals_const.csv`
  - `val_signals_const.csv`
- **Output directory:** `results_const_modified_mlp/`
- **Output files:**
  - `best_frequency_const_modified_mlp.pth`
  - `training_history_const_modified_mlp.csv`
  - `validation_results_const_modified_mlp.csv`
  - `training_validation_loss_const_modified_mlp.png`

3. 1D CNN: Constant-Frequency Signals
- **Script:** `train_validate_const_cnn.py`
- **Input data:**
  - `train_signals_const.csv`
  - `val_signals_const.csv`
- **Output directory:** `results_const_cnn/`
- **Output files:**
  - `best_frequency_const_cnn.pth`
  - `training_history_const_cnn.csv`
  - `validation_results_const_cnn.csv`
  - `training_validation_loss_const_cnn.png`

4. Original MLP: Time-Varying Frequencies
- **Script:** `train_validate_varying_mlp.py`
- **Input data:**
  - `train_signals_varying.csv`
  - `val_signals_varying.csv`
- **Output directory:** `results_varying_mlp/`
- **Output files:**
  - `best_frequency_varying_mlp.pth`
  - `training_history_varying_mlp.csv`
  - `validation_results_varying_mlp.csv`
  - `training_validation_loss_varying_mlp.png`
  - `frequency_predictions_examples_varying_mlp.png`

5. Modified MLP: Time-Varying Frequencies
- **Script:** `train_validate_varying_modified_mlp.py`
- **Input data:**
  - `train_signals_varying.csv`
  - `val_signals_varying.csv`
- **Output directory:** `results_varying_modified_mlp/`
- **Output files:**
  - `best_frequency_varying_modified_mlp.pth`
  - `training_history_varying_modified_mlp.csv`
  - `validation_results_varying_modified_mlp.csv`
  - `training_validation_loss_varying_modified_mlp.png`
  - `frequency_predictions_examples_varying_modified_m`

6. 1D CNN: Time-Varying Frequencies
- **Script:** `train_validate_varying_cnn.py`

- **Input data:**
  - `train_signals_varying.csv`
  - `val_signals_varying.csv`
- **Output directory:** `results_varying_cnn/`
- **Output files:**
  - `best_frequency_varying_cnn.pth`
  - `training_history_varying_cnn.csv`
  - `validation_results_varying_cnn.csv`
  - `training_validation_loss_varying_cnn.png`
  - `frequency_predictions_examples_varying_cnn.png`

D. Model Testing

- **Script:** `test_varying_three_models.py`
- **Models:** Original MLP, modified MLP, and 1D CNN
- **Test data:** `test_signals_varying.csv`
- **Output directory (synthetic data):** `test_results_varying/`
- **Output directory (real data):** `test_results_varying_real_data/`
- **Output files:**
  - `test_model_comparison.csv`
  - **Model subdirectories:** `MLP/`, `mMLP/`, `CNN/`
    - `test_frequency_examples.png`
    - `test_metrics_per_output.csv`
    - `test_predictions_varying.csv`